Minab School Attack: On 28 Feb. 2026, the first day of the US-Iran war, the Shajareh Tayyebeh, an elementary school in Minab, Hormozgan, Southern Iran, was hit by a Tomahawk missile, killing 156 civilians, including 120 schoolchildren, 35 adults, and a fetus. As of 22 Sept. 2026, the attack was the deadliest in civilian casualties in an ongoing war.



Outline: W05

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k-Nearest Neighbor

IIII) Evaluation

Metrics, Methodology

Fully Supervised Classification Evaluation

Technique (kNN, LR, SVM, …) does *not* matter …

Students (technique) learn from the course (labeled set) differently, yet all are evaluated/judged the same.



Statue of Lady Justice blindfolded and holding a balance and a sword, outside the Court of Final Appeal, Hong Kong

$$(o, c) \rightarrow f(o) \cong c$$

Lady Justice, or the goddess Justitia, an allegorical personification of justice and moral force in ancient Roman art known as Iustitia or Justitia. Her attributes are scales, a sword and blindfold.
https://en.wikipedia.org/wiki/Lady_Justice

Fully Supervised Boolean Classification Evaluation

Accuracy

$$(o, 1) \rightarrow f(o) \cong 1$$
$$(o, 0) \rightarrow f(o) \cong 0$$

For many (o, c), how often $f(o) = 1$ if $c = 1$, and how often $f(o) = 0$ if $c = 0$ …

Being in a binary choice (True/False) exam …

For a random (o, c), what's the probability

A random question from the pool of binary choice questions

$$\mathsf{P}(f(o) = c \mid (o, c)) = \mathsf{P}(f(o) = 1) \times \mathsf{P}(c = 1) + \mathsf{P}(f(o) = 0) \times \mathsf{P}(c = 0)$$

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Fully Supervised Boolean Classification Evaluation

Accuracy: **balanced** distribution of classes

$$(o, 1)_{001 \ldots 050} \rightarrow f(o) \cong 1$$
$$(o, 0)_{051 \ldots 100} \rightarrow f(o) \cong 0$$

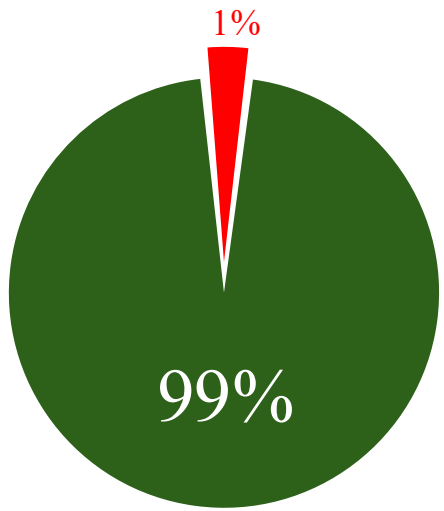
	Accuracy
A biased pessimist $f(o) = 1$ unless proved otherwise!	0.5
A biased optimist $f(o) = 0$ unless proved otherwise.	0.5
An unbiased drunken $f(o) = 1$ or 0 randomly, who cares!	0.5
An unbiased gambler $f(o) = 1$ or 0 by flip/toss a coin …	0.5
A data mining technique (method)	0.50000001 at least
The Oracle of Delphi	1.0

Fully Supervised Boolean Classification Evaluation

Accuracy: **Im**balanced (**un**balanced) distribution of classes

$$(o, 1)_{001 \ldots 001} \rightarrow f(o) \cong 1$$
$$(o, 0)_{002 \ldots 100} \rightarrow f(o) \cong 0$$

	Majority class	Minority class
Email spam detection	Not spam	Spam
Credit-card fraud detection	Legitimate	Fraudulent
Medical diagnosis	Healthy	Disease
Loan default prediction	Repays	Defaults
Customer churn	Stays	Churns
Network intrusion detection	Normal traffic	Attack
Manufacturing defect detection	No defect	Defective
Insurance claim fraud	Legitimate	Fraudulent
Ad click prediction	Does not click	Clicks
Fake review detection	Genuine	Fake
Cancer screening	Negative	Positive



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Fully Supervised Boolean Classification Evaluation

Accuracy: Imbalanced distribution of classes

$$(o, 1)_{001 \ldots 001} \rightarrow f(o) \cong 1$$
$$(o, 0)_{002 \ldots 100} \rightarrow f(o) \cong 0$$

	Accuracy
A biased pessimist $f(o) = 1$ unless proved otherwise!	0.01 (poor but savior)
A biased optimist $f(o) = 0$ unless proved otherwise.	0.99 (accurate but useless)
An unbiased drunken $f(o) = 1$ or 0 randomly, who cares!	0.50
An unbiased gambler $f(o) = 1$ or 0 by flip/toss a coin …	0.50
A data mining technique (method)	0.99000001 at least
The Oracle of Delphi	1.0

Fully Supervised Boolean Classification Evaluation

Accuracy: Imbalanced distribution of classes

$$(o, 1)_{001 \ldots 001} \rightarrow f(o) \cong 1$$
$$(o, 0)_{002 \ldots 100} \rightarrow f(o) \cong 0$$

A biased pessimist $f(o) = 1$ unless proved otherwise!
A biased optimist $f(o) = 0$ unless proved otherwise.

Accuracy
0.01 (poor but savior)
0.99 (accurate but useless)

Not a good metric for unbalanced classes; gives low score to a useful model and high score to a useless!
→ Because it gives equal importance to inequal classes!
→ We need to focus evaluating f for minority class …
→ Other metrics …



Outline: W05

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k-Nearest Neighbor

IIII) Evaluation

Confusion Matrix

Fully Supervised Boolean Classification Evaluation

https://en.wikipedia.org/wiki/Precision_and_recall

		True condition				
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$	
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$	
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR−}}$	F_1 score = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) = $\frac{\text{FNR}}{\text{TNR}}$		

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Fully Supervised Boolean Classification Evaluation

Contingency Table, Confusion Matrix

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR−}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) = $\frac{\text{FNR}}{\text{TNR}}$	F_1 score = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

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Fully Supervised Boolean Classification Evaluation

An Oracle, a pessimist, an optimist, a drunkard, a gambler

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive** ?	**False positive**, ?	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, ?	**True negative** ?	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR−}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) = $\frac{\text{FNR}}{\text{TNR}}$	F_1 score = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

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Fully Supervised Boolean Classification Evaluation

An Oracle, pessimist, optimist, drunkard, gambler in an **im**balanced dataset (1 vs. 99)

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	**True positive** ?	**False positive**, ?
	Predicted condition negative	**False negative**, ?	**True negative** ?

Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$

Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$

Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$

False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$

False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$

Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$

True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$

False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$

Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$

Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR−}}$

F_1 score = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$

Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$

Negative likelihood ratio (LR−) = $\frac{\text{FNR}}{\text{TNR}}$

Outline: W05

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k-Nearest Neighbor

IIII) Evaluation

Confusion Matrix → A Single Score: Precision

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Fully Supervised Boolean Classification Evaluation

Precision: How many of **the model's** catches (positive) are correct?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$

From the model's perspective ...

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Fully Supervised Boolean Classification Evaluation

Precision: How many of **the model's** catches (positive) are correct?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$

From the model's perspective in a balanced dataset (50 vs. 50) …

- Pessimist: $\frac{TP=50}{100} = 0.5$
- Optimist: Never says positive $\frac{TP=0}{0} \rightarrow 0.0$
- Drunk/Gambler: $\frac{TP=25}{50} = 0.5$

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Fully Supervised Boolean Classification Evaluation

Precision: How many of the model's catches (positive) are correct?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$

From the model's perspective in an imbalanced dataset (1 vs. 99) …

- Pessimist: $\frac{TP=1}{100} = 0.1$ (but a valuable model, catches that rare positive)
- Optimist: Never says positive $\frac{TP=0}{0} \rightarrow 0.0$
- Drunk/Gambler: $\frac{TP=?}{?} = ?$

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Fully Supervised Boolean Classification Evaluation

Precision: How many of the model's catches (positive) are correct?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$

From the model's perspective ...

What scenarios/applications require high precision?
Better question would be *"what happens if we have low precision? Will it be disaster?"*

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Fully Supervised Boolean Classification Evaluation

Precision: How many of **the model's** catches (positive) are correct?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$

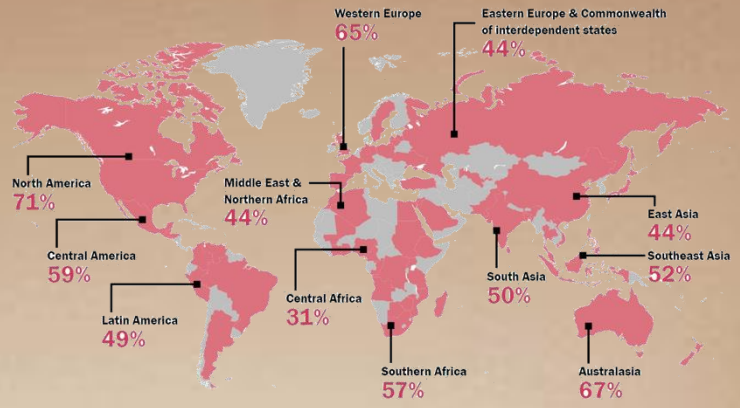
The observations are flying objects.
The classification task is catching/attacking enemy fighter jets.
The classifier is a Boolean missile system.
It infers (predicts) positive class*!* That is an enemy fighter jet approaching, fire*!*
From **the missile system** perspective, what happens if it fires with low precision?

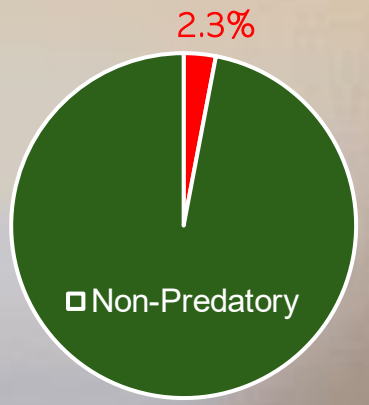


Ukraine International Airlines Flight 752

https://en.wikipedia.org/wiki/Ukraine_International_Airlines_Flight_752







Online Grooming

Osprey: A Reference Framework for Online Grooming Detection

https://github.com/fani-lab/Osprey

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Fully Supervised Boolean Classification Evaluation

Precision: How many of **the model's** catches (positive) are correct?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$

The observations are online users.
The classification task is catching a child abuser (predator).
The classifier is a Boolean alarm or notifier.
It infers (predicts) positive class! That is an abuser, notify child or parent!
From **the alarm perspective**, what happens if it notify with low precision?
Majority of the notifications are false. Not a disaster. We'll check users one by one.

Outline: W06C

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k-Nearest Neighbor

IIII) Evaluation

Confusion Matrix → A Single Score: Recall

Fully Supervised Boolean Classification Evaluation

Recall: How many of **the dataset's** positives are **remembered** by the model?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error			
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$			

From the **universe** perspective …

Fully Supervised Boolean Classification Evaluation

Recall: How many of **the dataset's** positives are **remembered** by the model?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error			
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$			

From the **universe** perspective …

Pessimist?
Optimist?
Drunk/Gambler?

Fully Supervised Boolean Classification Evaluation

Recall: How many of **the dataset's** positives are **remembered** by the model?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error			
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$			

From the **universe** perspective in **balanced** classes (50 vs. 50)

Pessimist: $\frac{TP=50}{50} = 1.0$ (not useful though!)

Optimist: $\frac{TP=0}{50} = 0.0$

Drunk/Gambler: $\frac{TP=25}{50} = 0.5$

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Fully Supervised Boolean Classification Evaluation

Recall: How many of **the dataset's** positives are **remembered** by the model?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error			
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$			

From the **universe** perspective in imbalanced classes (1 vs. 99)

Pessimist: $\frac{TP=1}{1} = 1.0$ (savior)

Optimist: $\frac{TP=0}{1} = 0.0$

Drunk/Gambler: $\frac{TP=?}{1} = ?$

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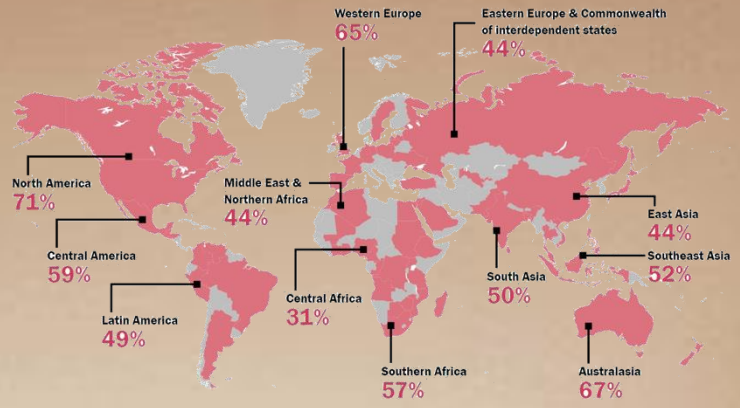
Fully Supervised Boolean Classification Evaluation

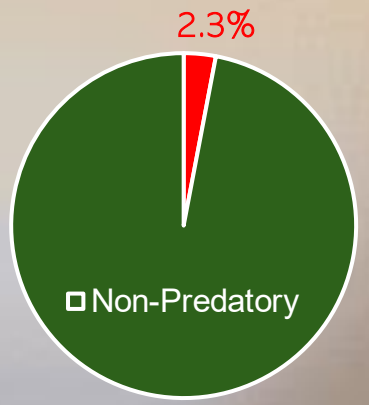
Recall: How many of **the dataset's** positives are **remembered** by the model?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error			
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$			

From the **universe** perspective …

What scenarios/applications require high recall?
Better question would be
"what happens if we have low recall? Will it be disaster?"





Online Grooming

Osprey: A Reference Framework for Online Grooming Detection

https://github.com/fani-lab/Osprey

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Fully Supervised Boolean Classification Evaluation

Recall: How many of the dataset's positives are remembered by the model?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error			
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$			

The observations are online users.
The classification task is catching a child abuser (predator).
The classifier is a Boolean alarm or notifier.
It infers positive class! That is an abuser, notify child or parent!

From the society perspective, what happens if it notify with low recall?
Cannot recall (remember) the predators. A disaster!

Fully Supervised Boolean Classification Evaluation

Recall: How many of **the dataset's** positives are **remembered** by the model?

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error			
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$			

The observations are flying objects.
The classification task is catching/attacking enemy fighter jets.
The classifier is a Boolean missile system.
There are fighting jets attacking a country.
From **the country's** perspective,
what happens if the missile system has low recall (forgetful)?



Atomic bombings of Hiroshima and Nagasaki: On 6 and 9 August 1945, the United States detonated two atomic bombs over the Japanese cities of Hiroshima and Nagasaki, respectively, during the final days of World War II. The aerial bombings killed 150,000 to 246,000 people, and remain the first and only uses of nuclear weapons in an armed conflict. For scaling reference, the population of the City of Victoria is 91,867.

Amiriyah Shelter Bombing: At 04:30 AM, 13 Feb 1991, two F-117 stealth bombers each dropped a 900-kilogram (2,000 lb) GBU-27 laser-guided bomb on the shelter. The first bomb cut through 3 metres (10 ft) of reinforced concrete before a time-delayed fuse exploded. Minutes later, the second bomb followed the path cut by the first bomb. Neighborhood residents heard screams as people tried to get out of the shelter. They screamed for four minutes. After the second bomb hit, the screaming ceased. More than 408 people were killed.



Minab School Attack: On 28 Feb. 2026, the first day of the US-Iran war, the Shajareh Tayyebeh, an elementary school in Minab, Hormozgan, Southern Iran, was hit by a Tomahawk missile, killing 156 civilians, including 120 schoolchildren, 35 adults, and a fetus. As of 22 Sept. 2026, the attack was the deadliest in civilian casualties in an ongoing war.



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Fully Supervised Boolean Classification Evaluation

Trade-off between Recall vs. Precision

		True condition			
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error			
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$			

The observations are flying objects.
The classification task is catching/attacking enemy fighter jets.
The classifier is a Boolean missile system.
There are fighting jets attacking a country.
High precision & high recall system is expensive, or unavailable!
(A) High precision but low recall?
(B) Low precision but high recall?

Fully Supervised Boolean Classification Evaluation

The observations are flying objects.
The classification task is catching/attacking enemy fighter jets.
The classifier is a Boolean missile system.
There are fighting jets attacking a country.
Trade-off between precision and recall …

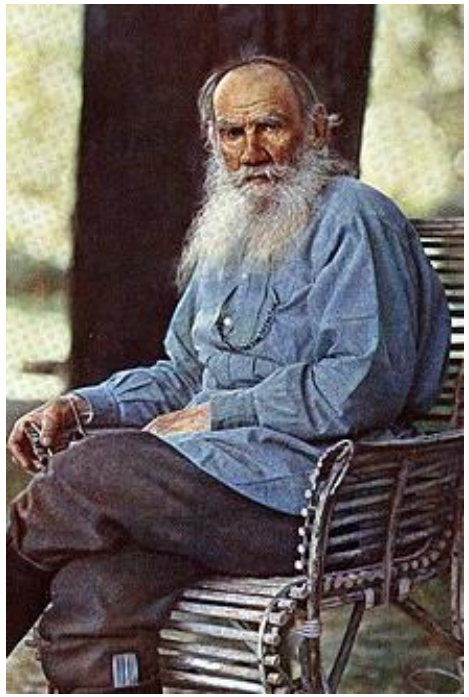
(A) High precision but low recall?
With high precision, we don't shot (our own) passengers.
With low recall, the model forgets an enemy fighter jet. It flies & bombs people. **But it is enemy**.

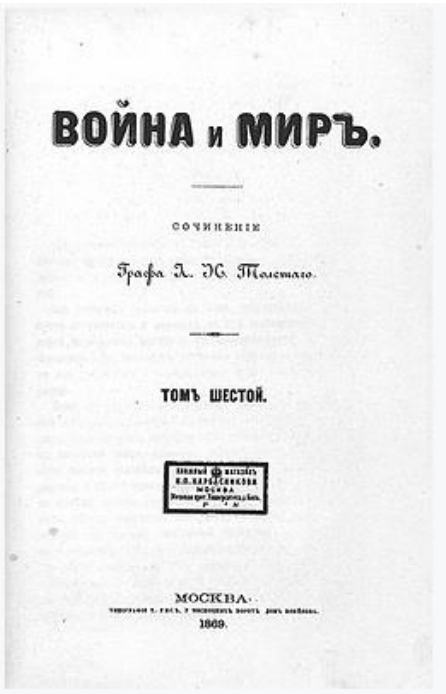
(B) Lower precision but higher recall?
With low precision, we shot (our own) passengers*!* But worth it as it might be fighter jet.
With high recall, the model remembers all enemy fighter jet. No bombing would happen.

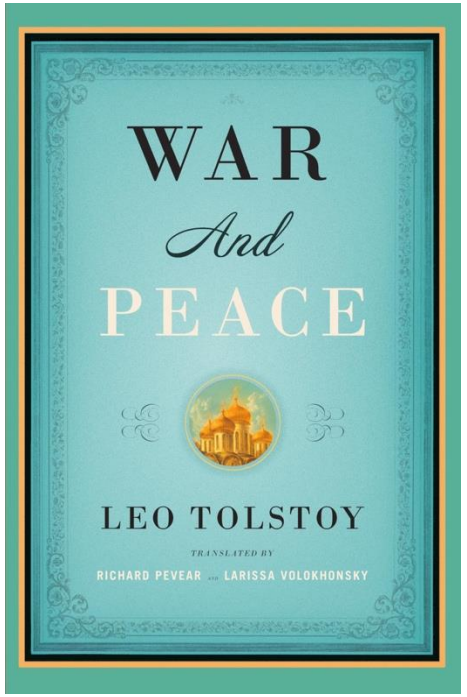
Fully Supervised Boolean Classification Evaluation

Are we at war or peace? *[fill in the blank and justify your answer]*

- At peace, we're optimist. We need [ ] missile system. Because …
- At war, we're pessimist. We need [ ] missile system. Because …







Count Lev Nikolayevich Tolstoy /ˈtoʊlstɔɪ, ˈtɒl-/
1828 – 1910, Russian
In English, also referred to as Leo Tolstoy
One of the greatest and most influential authors of all time.
War and Peace, is his epic novel.

Outline: W06C

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k-Nearest Neighbor

IIII) Evaluation

Confusion Matrix → A Single Score from Both Views

F-Measure

Fully Supervised Boolean Classification Evaluation

A Single Score: Mean (Average) of Precision and Recall

$$\text{AVG-PR} = \frac{\text{P}+\text{R}}{2}$$

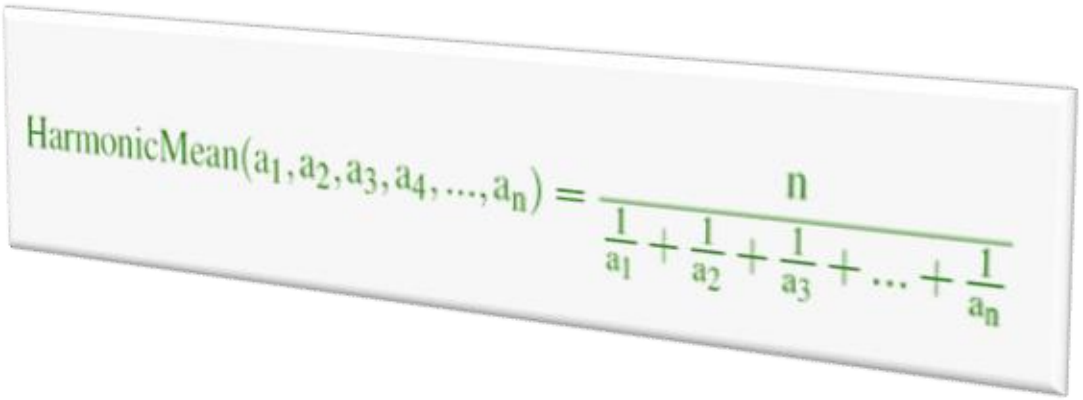
Equal weights to Precision and Recall …

Equality is not fair!

High precision may discount low recall, or vice versa!

Fully Supervised Boolean Classification Evaluation

A Single Score: Harmonic Mean (Average) of Precision and Recall



$$\text{Harmonic AVG-PR} = \frac{2}{\frac{1}{P}+\frac{1}{R}} = 2\times\frac{P\times R}{P+R}$$

Still equal weights to Precision and Recall and equality is not fair!

But at least conservative! More toward the lower number.

If $P \rightarrow 1$, Harmonic AVG-PR $= \frac{2R}{1+R}$

If $R \rightarrow 1$, Harmonic AVG-PR $= \frac{2P}{P+1}$

Fully Supervised Boolean Classification Evaluation

A Single Score: Weighted Harmonic Mean of Precision and Recall

$$F_\alpha = \frac{2}{\alpha\frac{1}{P} + (1-\alpha)\frac{1}{R}}$$

But again, in academia, we like making things complicated to look professional …

Fully Supervised Boolean Classification Evaluation

A Single Score: Weighted Harmonic Mean of Precision and Recall

$$F = \frac{1}{\alpha\frac{1}{P} + (1-\alpha)\frac{1}{R}} \quad \text{or} \left(\text{with } \beta^2 = \frac{1-\alpha}{\alpha}\right)$$

$$F_\beta = \frac{(\beta^2+1)PR}{\beta^2 P + R} \begin{cases} \beta > 1\text{: favors Recall} & \text{If } \beta \to +\infty, \text{ F} = \text{R} \\ \beta = 1\text{: } \quad 2\dfrac{\text{P} \times \text{R}}{\text{P} + \text{R}} & \\ 0 \leq \beta < 1\text{: favors Precision} & \text{If } \beta \to 0.0, \text{ F} = \text{P} \end{cases}$$

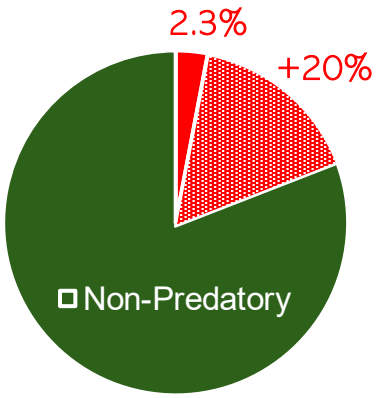
Fully Supervised Boolean Classification Evaluation

A Single Score: Weighted Harmonic Mean of Precision and Recall

Enhancing Online Grooming Detection via Backtranslation Augmentation*

https://hosseinfani.github.io/res/papers/2025_COLING_Enhancing_Online_Grooming_Detection_via_Backtranslation_Augmentation.pdf

original message	language: translation	backtranslation
'having it with minor'	French: *'l'avoir avec mineur'*	*'having sex with a minor'*
'i feel little aroused'	German: *'ich fühle mich ein wenig erregt'*	*'i'm feeling a little turned on'*
'you ever hooked up with anybody...?'	Chinese: '你有没有和网上的人交过?'	*'have you ever had sex with ...?'*
'like two guys doing each other?'	German: *'wie zwei typen, die es miteinander treiben?'*	*'like two guys having sex?'*



			Δf0.5			Δf1			Δf2		
			google	m2m100	nllb	google	m2m100	nllb	google	m2m100	nllb
gru	none		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	+fr		+8.76	+2.61	+2.65	+7.08	+2.84	+2.63	+5.80	+2.46	+3.35
	+fa		+1.68	+3.50	+4.05	+1.88	+3.64	+3.39	+2.43	+5.42	+3.04
	+de		+4.10	+3.81	+9.00	+3.76	+2.97	+6.39	+4.22	+2.92	+2.27
	+zh		-8.97	+1.55	+5.63	-5.87	+2.14	+4.27	-3.92	+4.89	+2.61
	+ca		+7.29	**+7.44**	+11.03	+6.73	+5.40	+8.04	+5.01	+2.67	+4.70
	+ps		9.57	-5.32	+13.60	+6.48	-3.79	+7.58	+3.04	+1.56	+0.52
	+is		+3.73	-2.24	+10.14	+4.04	-0.68	+6.98	+6.27	+2.25	+3.24
	+my		**+12.51**	-3.13	+9.34	+9.13	-1.66	+6.06	+4.63	+1.46	+2.00
	+fr+ca	Western Romance	+8.07	-4.40	**+15.29**	+7.02	-2.35	**+9.94**	+6.76	-0.81	+3.52
	+fa+ps	Iranic	-3.31	-6.71	+2.91	-1.88	-4.75	+2.62	+0.34	+0.78	+3.60
	+de+is	West Germanic	+11.84	+5.94	+9.43	**+9.39**	**+5.60**	+7.65	**+6.98**	**+6.63**	**+6.16**
	+zh+my	Sino-Tibetan	+8.83	-8.76	-3.15	+6.80	-6.33	-1.21	+4.55	-7.05	+0.66
	+fr+fa+de+zh	high-resource	+5.88	-1.36	+10.14	+4.62	-0.40	+7.63	+3.94	+3.01	+4.97
	+ca+ps+is+my	low-resource	+8.10	-1.38	+10.41	+6.61	-0.52	+5.26	+5.63	+2.64	-1.00
	all		+4.25	-0.32	+6.51	+4.30	+0.06	+5.20	+6.02	+2.16	+4.64

Outline: W06C

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k-Nearest Neighbor

IIII) Evaluation

Confusion Matrix → Multiclass

Fully Supervised Multiclass Classification Evaluation

		Truth	
		Like	Neutral
Model	Like	8	1
	Neutral	3	100

https://www.wsj.com/video/facebook-users-outraged-over-emotion-experiment/AFB5F0DF-3AB3-4E96-B2E1-1FA95D56A786



Facebook apologises for psychological experiments on users

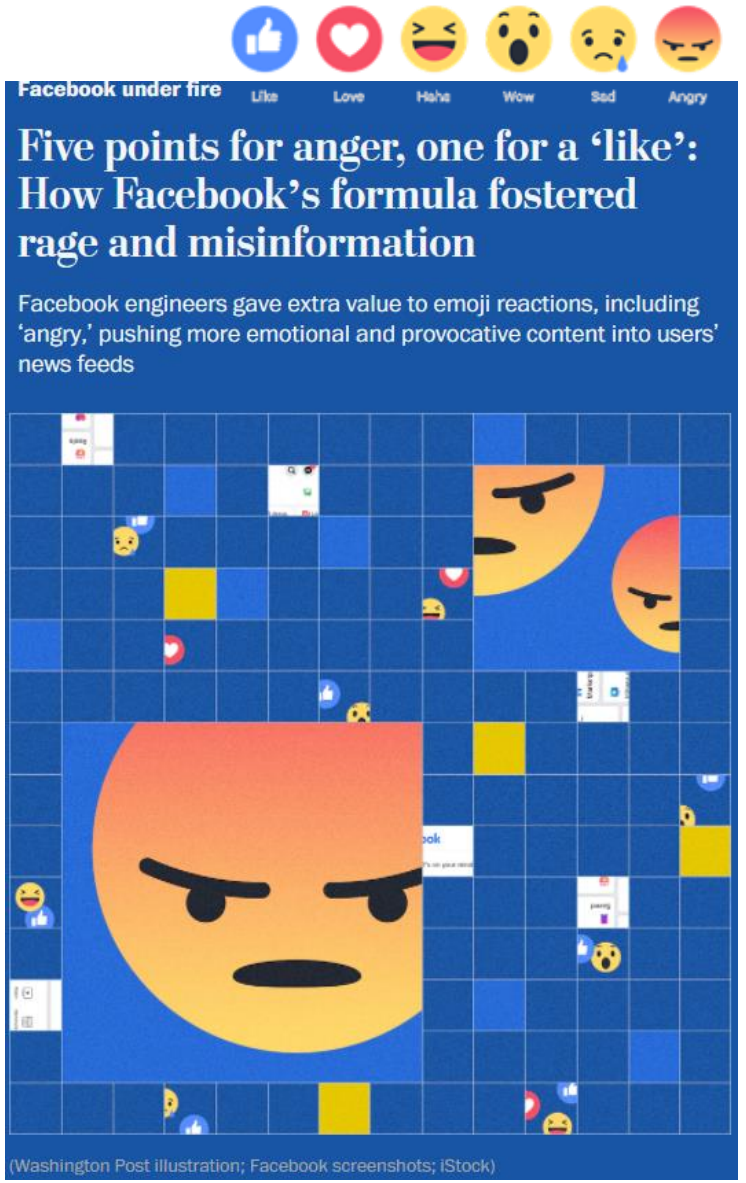
The second most powerful executive at the company, Sheryl Sandberg, says experiments were 'poorly communicated'



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Fully Supervised Multiclass Classification Evaluation

		Truth		
		Happy	Sad	Neutral
Model	Happy	8	10	1
	Sad	5	60	50
	Neutral	3	30	100



For 7 years, users begged for a **dislike** button. But Facebook **vehemently** resisted and feared a binary dislike button would introduce **cyberbullying, create a toxic/negative environment, or discourage people from posting entirely**.
https://www.nbcnews.com/tech/tech-news/emoji-reactions-cute-addition-facebook-became-headache-rcna3747
https://variety.com/2015/digital/news/facebook-dislike-sadness-sympathy-buttons-1201594213/
https://www.washingtonpost.com/technology/2021/10/26/facebook-angry-emoji-algorithm/